EXPVID: A BENCHMARK FOR EXPERIMENT VIDEO UNDERSTANDING & REASONING

Yicheng Xu[1], Yue Wu[1], Jiashuo Yu[1], Ziang Yan[1], Tianxiang Jiang[1], Yinan He[1], Qingsong Zhao[1], Kai Chen[1], Yu Qiao[1], Limin Wang[2], Manabu Okumura[3], Yi Wang[1]

[1]Shanghai AI Laboratory [2]Nanjing University [3]Tokyo Institute of Technology

ABSTRACT

Real-world laboratory experiments demand fine-grained visual perception, step-by-step procedural reasoning, and high-order scientific inference—capabilities that existing multimodal large language model (MLLM) benchmarks rarely test together. We introduce ExpVid, the first benchmark specifically designed to evaluate MLLMs on authentic, peer-reviewed wet-lab experiment videos. ExpVid organizes 10 tasks into a three-level hierarchy: *Level-1* probes fine-grained visual perception (tool, material, quantity, and operation recognition) on short action clips ($\sim$8 s); *Level-2* assesses procedural understanding (step ordering, sequence generation, completeness verification, and step prediction) on stage-level segments ($\sim$48 s); and *Level-3* requires full-experiment scientific reasoning (experimental analysis and scientific discovery) via cloze tasks over complete videos ($\sim$8 min). The benchmark is built from 390 peer-reviewed JoVE [JoVE] experiment videos spanning 13 scientific disciplines, yielding 7,800 vision-centric QA pairs. We evaluate 19 state-of-the-art MLLMs in both standard and chain-of-thought ("think") modes. Results reveal substantial gaps: GPT-5 [OpenAI, 2025] achieves the highest Level-2 average (57.5) and Level-3 average (56.4); Gemini-2.5-Flash [DeepMind, 2025a] leads Level-1 (60.2); and the best open-source model, InternVL3-78B [Zhu et al.], reaches 50.9/41.9/37.7 across the three levels. Removing visual frames sharply degrades performance, confirming the vision-centric design. ExpVid provides a rigorous testbed for building MLLMs that can perceive, verify, and reason about real laboratory workflows.

AI-Generated

1 Introduction

Understanding laboratory experiments from video is a fundamental capability for artificial intelligence systems that aim to assist scientific research. Wet-lab experiments involve intricate manual procedures, specialized equipment, and domain-specific reasoning that together pose unique challenges for multimodal large language models (MLLMs). An AI system that can parse experiment videos must not only recognize fine-grained actions and apparatus but also understand multi-step protocols and draw scientific inferences from observed outcomes. Such capability would enable applications ranging from automated lab assistants to scalable quality control in experimental reproducibility.

Recent years have seen rapid progress in video understanding benchmarks. General-purpose benchmarks such as MVBench [Li et al., 2024] and Video-MME [Fu et al., 2025] evaluate broad perceptual abilities across diverse video genres, while MLVU [Zhou et al., 2024] and VR-Bench [Yu et al., 2025] target long-form video comprehension and visual reasoning, respectively. On the scientific knowledge front, MMMU [Yue et al., 2024], MMVU [Zhao et al., 2025], and Video-MMMU [Hu et al., 2025] assess expert-level understanding of academic content, and ScienceQA [Saikh et al., 2022], SciBench [Wang et al., 2023], and MMStar [Chen et al., 2024] probe various dimensions of scientific problem solving. However, these benchmarks share a common limitation: they either focus on generic everyday activities or test static scientific knowledge, overlooking the fine-grained, long-horizon, and inherently procedural nature of real wet-lab experiments. None of them requires a model to jointly perceive apparatus manipulations, reconstruct multi-step experimental protocols, and reason about scientific outcomes—the three core competencies that define experimental literacy.

To bridge this gap, we introduce ExpVid, a comprehensive benchmark for **Exp**eriment **Vid**eo understanding and reasoning. ExpVid is built on 390 professionally produced experiment videos from the Journal of Visualized Experiments (JoVE) [JoVE], spanning 13 scientific disciplines with an average video duration of 489 seconds. The benchmark comprises 7,800 question–answer pairs organized into 10 distinct tasks across a three-level cognitive hierarchy that mirrors the scientific workflow: **Level-1 (Perception)** tests fine-grained visual recognition through multiple-choice

questions about equipment, samples, actions, and phenomena; **Level-2 (Procedure)** evaluates understanding of experimental protocols, including step ordering, temporal grounding, and causal linking of procedural steps; and **Level-3 (Reasoning)** probes higher-order scientific reasoning through cloze-style tasks that require predicting experimental outcomes, identifying underlying principles, and assessing safety or failure modes. All annotations are vision-centric, carefully designed to prevent models from shortcutting via text priors or memorized scientific knowledge without actually watching the video.

We evaluate 19 MLLMs on ExpVid, including both open-source and closed-source systems. Our evaluation reveals several key findings: closed-source models consistently outperform their open-source counterparts; GPT-5 achieves the highest scores on Level-2 and Level-3 tasks; Gemini-2.5-Flash leads on Level-1 perceptual tasks; and the best-performing open-source model is InternVL3-78B. Crucially, ablation studies confirm that visual input is essential—text-only baselines and Scientific Foundation Encyclopedias (SFE) [Zhou et al., 2025] alone are insufficient for solving ExpVid tasks, validating our vision-centric design.

Our contributions can be summarized as follows:

- We identify a critical gap in existing MLLM benchmarks: the absence of evaluations targeting authentic, long-form wet-lab experiment videos that require integrated perception, procedural understanding, and scientific reasoning.
- We propose ExpVid, a three-level hierarchical benchmark comprising 390 JoVE experiment videos across 13 disciplines, with 7,800 vision-centric QA pairs spanning 10 tasks designed to mirror the scientific workflow from observation through reasoning.
- We conduct a comprehensive evaluation of 19 state-of-the-art MLLMs and provide detailed analyses across task types, disciplines, and model families, revealing significant performance gaps that highlight open challenges.
- We demonstrate through ablation studies that visual grounding is indispensable for ExpVid, confirming that our benchmark resists text-based shortcutting and genuinely measures video understanding.

2 Related Work

2.1 Video Understanding Benchmarks

Video understanding has been extensively studied through benchmarks that target a range of capabilities. Early efforts such as ActivityNet [Caba Heilbron et al., 2015] and Charades [Sigurdsson et al., 2016] established standard evaluation protocols for action recognition and temporal activity detection in everyday scenarios. More recently, MVBench [Li et al., 2024] introduced a systematic evaluation of temporal understanding across 20 tasks, while Video-MME [Fu et al., 2025] provided a multi-modal evaluation pipeline covering short to long videos. MLVU [Zhou et al., 2024] specifically addresses long-video understanding with tasks requiring comprehension over extended temporal horizons, and EgoSchema [Mangalam et al., 2023] focuses on egocentric video reasoning over lengthy first-person recordings. Temporal grounding benchmarks [Lei et al., 2021] further evaluate the ability to localize events within videos. VR-Bench [Yu et al., 2025] targets visual reasoning capabilities that go beyond recognition. Despite this progress, existing video benchmarks predominantly feature everyday activities, entertainment, or egocentric recordings. They do not capture the specialized visual vocabulary, extended procedural structure, or domain-specific reasoning demands of scientific experiments, motivating the development of ExpVid.

2.2 Scientific and Knowledge Benchmarks

A parallel line of work has developed benchmarks for scientific and expert-level knowledge assessment. MMMU [Yue et al., 2024] evaluates multimodal understanding across 30 academic subjects using college-level exam questions, and its video extension Video-MMMU [Hu et al., 2025] adapts this paradigm to video inputs. MMVU [Zhao et al., 2025] focuses on expert-level visual understanding in specialized domains. On the text-centric side, ScienceQA [Saikh et al., 2022] and SciBench [Wang et al., 2023] test scientific problem-solving across multiple disciplines. MathVista [Lu et al., 2023] targets mathematical reasoning with visual contexts, while MMStar [Chen et al., 2024] provides a large-scale evaluation that specifically guards against visual shortcutting. SFE [Zhou et al., 2025] and MathVision [Wang et al., 2024b] further extend scientific evaluation to perception-reasoning and mathematical visual reasoning settings, respectively. While these benchmarks assess factual scientific knowledge, they do not evaluate the ability to understand how experiments are physically conducted—the procedural and observational skills that ExpVid is specifically designed to measure.

AI-Generated

2.3 Procedural and Instructional Video Understanding

Understanding procedural and instructional videos requires parsing ordered sequences of actions, identifying tools and materials, and reasoning about goals and dependencies. Prior work has largely focused on cooking and household domains: the TACoS benchmark [Das et al., 2013] pairs cooking videos with natural language descriptions for grounded language understanding. More broadly, instructional video datasets have been developed for how-to activities [Gabrieli et al., 2025]. Recently, attention has turned toward scientific procedures: SciVid [Hasson et al., 2025] curates experiment videos for scientific video understanding, and FineBio [Yagi et al., 2025] provides fine-grained annotations of biology laboratory protocols. However, these efforts either cover a narrow set of disciplines or focus exclusively on action segmentation without testing higher-order scientific reasoning. ExpVid addresses these limitations by spanning 13 disciplines and organizing evaluation into a three-level hierarchy that progresses from perceptual recognition through procedural comprehension to scientific reasoning, providing a holistic assessment of experiment video understanding.

3 ExpVid: A Scientific Experiment Video Benchmark

3.1 Overview

ExpVid evaluates MLLMs across the full spectrum of competencies required to understand a wet-lab experiment: recognizing the right tools and materials, verifying that individual steps are executed correctly, and drawing scientifically grounded conclusions from the complete procedure. To this end, we define a three-level task hierarchy illustrated in Figure 1:

- **Level-1 (Perception):** Four multiple-choice recognition tasks on short action-level clips ($\sim$8 s): Tool Recognition, Material Recognition, Quantity Recognition, and Operation Recognition.
- **Level-2 (Procedure):** Four procedural tasks on stage-level segments ($\sim$48 s): Step Ordering, Sequence Generation, Completeness Verification, and Step Prediction.
- **Level-3 (Reasoning):** Two open-ended cloze tasks over full experiment videos ($\sim$8 min): Experimental Analysis and Scientific Discovery.

In total, ExpVid provides 7,800 QA pairs from 390 experiments, with $\sim$10k action clips for Level-1 and $\sim$3.5k stage segments for Level-2.

3.2 Data Collection and Curation

Source. We source videos from the Journal of Visualized Experiments (JoVE) [JoVE], a peer-reviewed publication dedicated to video-based scientific communication. JoVE videos come with paired ASR transcripts, structured protocols, and corresponding research articles, providing multi-modal grounding for annotation. We focus on the "JoVE Research" category, which features complete experimental workflows.

Quality Filtering. Raw candidate videos are filtered using DeepSeek-R1 [Guo et al., 2025a] across five dimensions: Continuity, Alignment, Clarity, Integrity, and Focus. Videos must achieve an overall score $\geq$4.0 with no individual dimension scoring below 3.5. We further restrict video durations to the 25th–75th percentile range (378–728 s) to exclude extremely short or unwieldy videos.

Discipline Curation. Expert curators select up to 30 videos per scientific discipline, yielding a final corpus of **390 videos spanning 13 disciplines** (including biology, chemistry, neuroscience, medicine, and biochemistry) with an average duration of 489 s. The construction pipeline is illustrated in Figure 2.

3.3 Data Preprocessing

ExpVid operates at three temporal scales, each requiring distinct preprocessing.

Level-1 (Action Clips). We segment each video into short clips by aligning ASR sentence boundaries with the video timeline, producing $\sim$10k clips averaging $\sim$8 s each, where each clip captures a single atomic action.

Level-2 (Stage Segments). We use DeepSeek-R1 [Guo et al., 2025a] to identify stage boundaries (targeting 20–60 s per segment) and extract a sequential list of experimental steps from the transcript, yielding $\sim$3.5k stage segments averaging $\sim$48 s.

AI-Generated

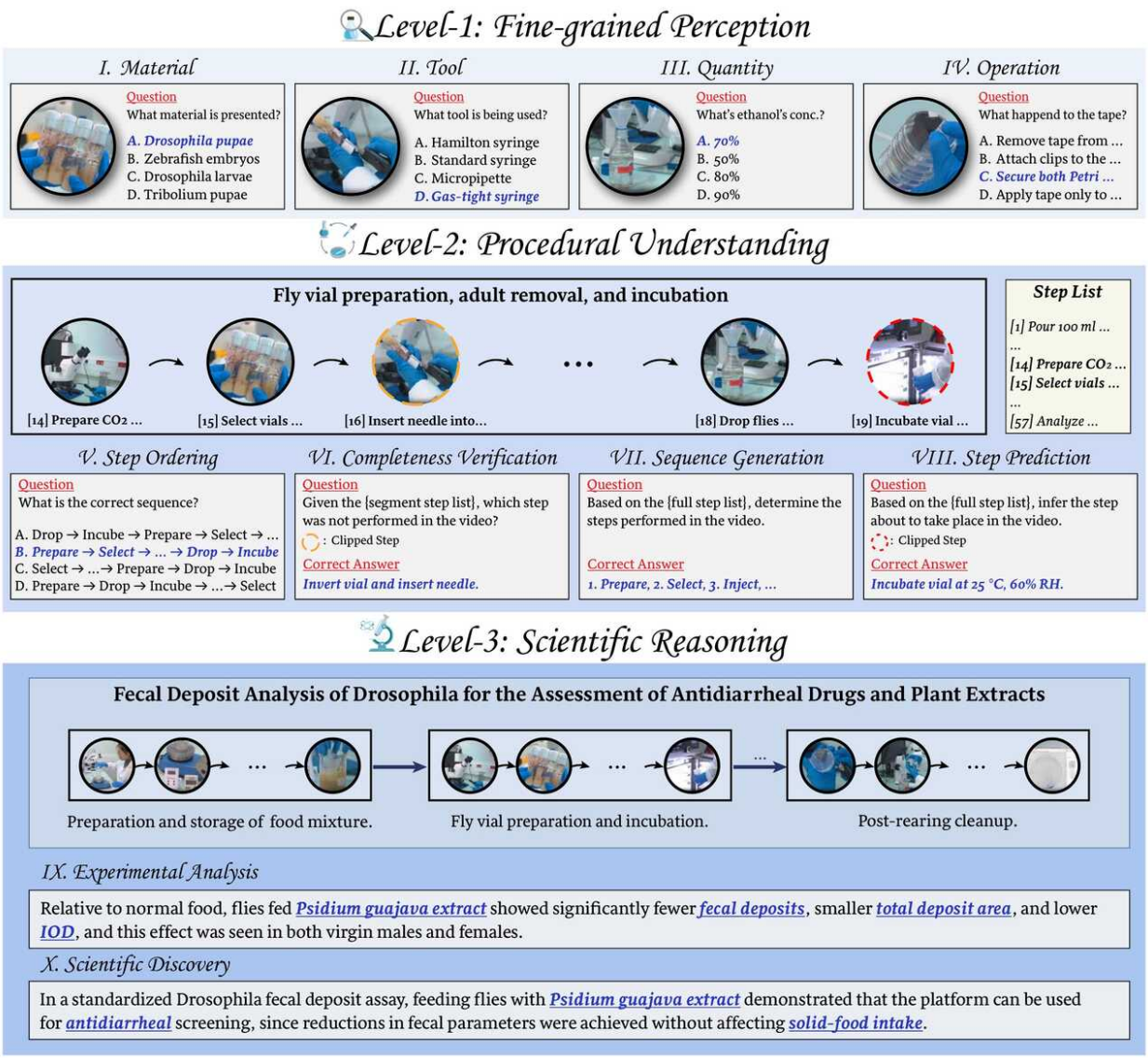

Figure 1: Illustration of the three-level task hierarchy in ExpVid. Level-1 tasks probe fine-grained visual perception on short action clips; Level-2 tasks assess procedural understanding on stage segments; Level-3 tasks require full-experiment scientific reasoning over complete videos.

Level-3 (Full Experiments). The full procedure video is retained for each experiment. Concluding slides, summary screens, and data-analysis segments are removed to prevent models from answering Level-3 questions by reading on-screen text rather than reasoning from visual observations.

3.4 Task Definition and Annotation

All annotations follow a *vision-centric* design: questions are crafted so that the correct answer requires observing the video frames, not merely reading the transcript or relying on language priors.

Level-1 Tasks. For each action clip, DeepSeek-R1 [Guo et al., 2025a] extracts candidate targets from the transcript; Qwen2.5-VL [Bai et al., 2025b] generates visual captions to confirm that the target is visually present; and DeepSeek-R1 crafts four-option MCQs with plausible distractors. Specific distractor strategies are:

- **Tool Recognition:** Distractors are visually or functionally similar instruments.

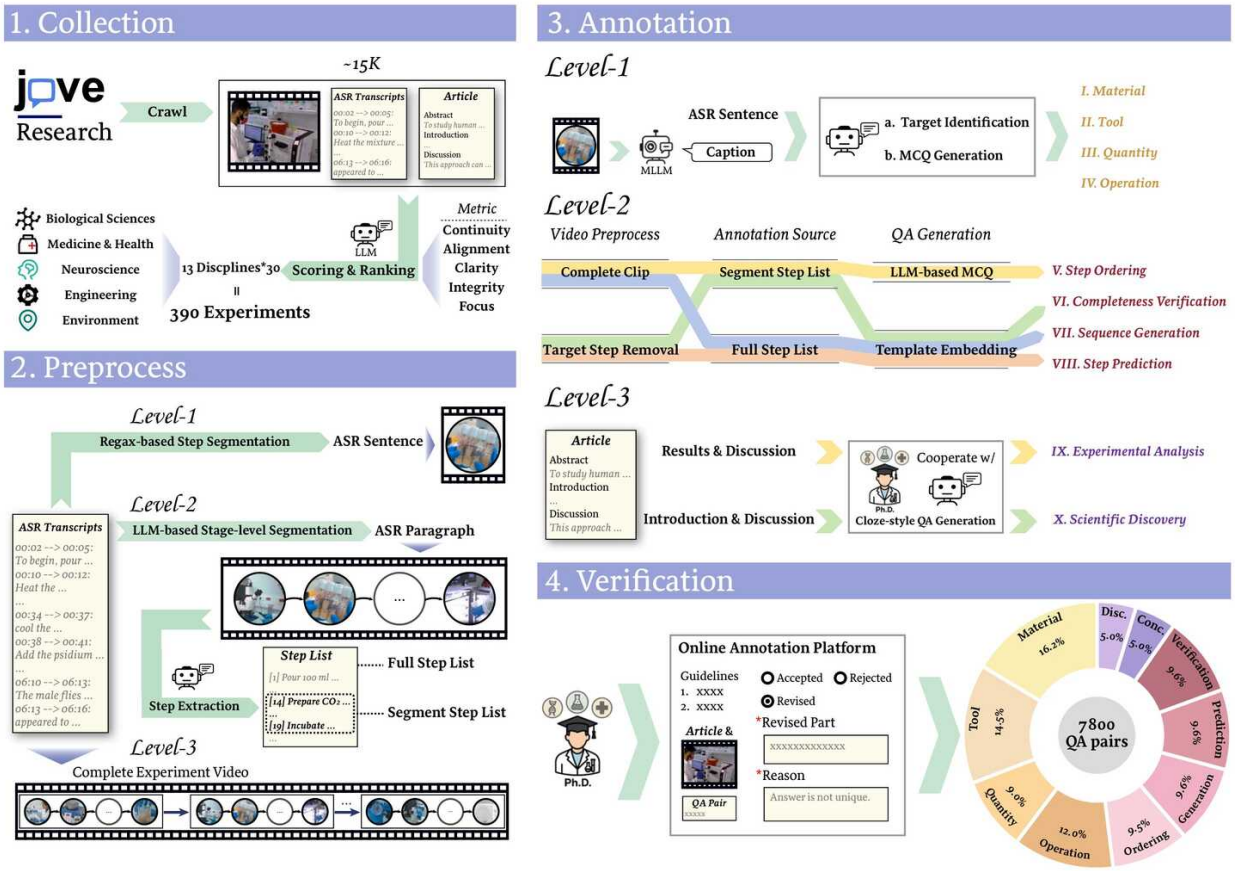

Figure 2: Overview of the ExpVid construction pipeline, covering video filtering, multi-level preprocessing, LLM-assisted annotation, and expert quality control.

- **Material Recognition:** Distractors have similar appearance or chemical function.
- **Quantity Recognition:** Distractors are near-range numeric alternatives of the same unit.
- **Operation Recognition:** Distractors are visually confusable actions.

Level-2 Tasks. Level-2 annotations are built from the step lists extracted during preprocessing. DeepSeek-R1 [Guo et al., 2025a] generates plausible permutations, candidate step pools, and missing-step variants:

- **Step Ordering:** Given shuffled steps from a procedure segment, place them in correct experimental order.
- **Sequence Generation:** Select the complete set of steps appearing in a segment from a candidate pool that includes plausible but absent steps. Evaluated with the Jaccard index.
- **Completeness Verification:** Determine whether a presented sequence of steps is complete or has a missing step (binary task).
- **Step Prediction:** Given observed steps in a segment, predict the most likely next experimental step.

Level-3 Tasks. Level-3 annotation draws on each video's paired research paper. Paper sections are extracted with MinerU [Wang et al., 2024a]; GPT-5 [OpenAI, 2025] summarizes key experimental anchors; and PhD-level domain experts (in biology, chemistry, and medicine) author cloze questions that require full-video reasoning and cannot be answered from the transcript alone:

- **Experimental Analysis:** Fill-in-the-blank questions about experimental conditions, controls, or observed phenomena requiring integration of visual evidence across the full video.
- **Scientific Discovery:** Fill-in-the-blank questions about key findings or conclusions, requiring both procedural understanding and scientific reasoning.

Table 1: Comparison between ExpVid and some MLLM benchmarks. A and M indicate automatic and manual annotation, respectively.

Benchmark	#QA Pairs	#Videos	Avg. Sec.	#Task Types	Annotation	Domain
General Video Benchmarks						
MVBench [Li et al., 2024]	4,000	3,641	16.0	20	A+M	General
Video-MME [Fu et al., 2025]	2,700	900	1,017.9	1	M	General
MLVU [Zhou et al., 2024]	3,102	1,730	930.0	9	M	Narrative
VRBench [Yu et al., 2025]	9,468	960	5,796.0	1	M	Narrative
Knowledge-driven Benchmarks						
MMVU [Zhao et al., 2025]	3,000	1,529	51.4	2	M	Multi-disc.
Video-MMMU [Hu et al., 2025]	900	300	506.2	3	M	Multi-disc.
MathVision [Wang et al., 2024b]	3,040	–	–	1	M	Math
MathVista [Lu et al., 2023]	6,141	–	–	31	M	Math
MMMU [Yue et al., 2024]	11,500	–	–	2	M	Multi-disc.
ScienceQA [Saikh et al., 2022]	21,208	–	–	1	M	Science
SciBench [Wang et al., 2023]	789	–	–	1	M	Science
MMStar [Chen et al., 2024]	1,500	–	–	6	M	Multi-disc.
SFE [Zhou et al., 2025]	830	–	–	66	M	Science
ExpVid	7,800	390	489.0	10	A+M	Science

3.5 Quality Control and Expert Verification

All annotations undergo iterative expert verification by PhD-level annotators with domain expertise in biology, chemistry, and medicine. Annotators validate: (1) *visual visibility*—the annotated target is clearly present in the clip; (2) *distractor plausibility*—incorrect options are plausible but distinguishable upon careful viewing; (3) *step–segment alignment*—Level-2 labels correctly correspond to their associated segment; and (4) *answer uniqueness*—each question has exactly one correct answer. Questions failing any criterion are revised or discarded. Average annotation times are 0.3 h/question (Level-1), 0.5 h/question (Level-2), and 1.2 h/question (Level-3).

3.6 Evaluation Protocol

- **Level-1 (MCQ):** Top-1 accuracy—the fraction of questions for which the model's top-ranked option matches the ground truth.
- **Level-2 (Mixed):** Top-1 accuracy for Step Ordering (exact permutation match), binary accuracy for Completeness Verification, and Top-1 accuracy for Step Prediction. For Sequence Generation, we use the Jaccard index (set intersection over union) between the predicted and ground-truth step sets.
- **Level-3 (Cloze):** Per-blank accuracy, where each predicted fill is judged by Phi-3-mini [Abdin et al., 2024] against the reference answer. Scores are averaged across all blanks and then across all questions.

4 Experiments

4.1 Experimental Setup

We evaluate 19 MLLMs spanning a wide range of architectures, scales, and training paradigms. *Open-source models* include: Qwen2.5-VL-7B-Instruct and Qwen2.5-VL-72B-Instruct [Bai et al., 2025b], MiMo-VL-7B-RL, InternVL3-8B and InternVL3-78B [Zhu et al.], InternVL3.5-8B and InternVL3.5-38B [Wang et al., 2025], Intern-S1-mini and Intern-S1 [Bai et al., 2025a], Keye-VL-8B-Preview and Keye-VL-1.5-8B, GLM-4.1V-9B and GLM-4.5V [Hong et al., 2025], and Kimi-VL-A3B-Thinking [Team et al., 2025]. *Closed-source models* include: Seed-VL-1.5 [Guo et al., 2025b], Claude-Sonnet-4 [Anthropic, 2025], Gemini-2.5-Flash [DeepMind, 2025a], Gemini-2.5-Pro [DeepMind, 2025b], and GPT-5 [OpenAI, 2025].

Where supported, models are evaluated in both *standard* (non-think) and *think* (chain-of-thought reasoning) modes. Video frames are uniformly sampled; the number of frames follows each model's recommended or maximum supported context. Human baselines on Level-1 and Level-2 are established using domain-naive annotators.

AI-Generated

4.2 Main Results

Full results are reported in Table 2. We highlight the key observations below.

Closed-source models dominate. Frontier closed-source models consistently outperform open-source alternatives across all three levels. GPT-5 achieves the highest Level-2 average (57.5) and Level-3 average (56.4), with particularly strong performance on Sequence Generation (66.9) and Step Prediction (51.8). Gemini-2.5-Flash in think mode leads Level-1 (60.2), followed closely by Gemini-2.5-Pro without think (59.2). The best open-source model is InternVL3-78B, reaching 50.9 on Level-1, 41.9 on Level-2, and 37.7 on Level-3—revealing a substantial gap relative to frontier models.

Human baselines reveal calibration. Human annotators achieve a Level-1 average of 37.6 and a Level-2 average of 42.1, scores competitive with or exceeding many MLLMs. The relatively low human Level-1 score reflects the genuine difficulty of fine-grained visual recognition for domain-naive participants, while the Level-2 score indicates that general procedural intuition transfers across disciplines. These results confirm that ExpVid is well-calibrated: difficult but not intractable.

Procedural asymmetry. Models perform substantially better on Step Ordering—where InternVL3-78B reaches 87.1 and even mid-range models exceed 60–80—than on Sequence Generation (GPT-5 best at 66.9) or Step Prediction, which is near-zero for many models (e.g., Qwen2.5-VL-72B: 0.3). This reveals a fundamental asymmetry: MLLMs can rearrange a given set of steps but struggle to generate or predict procedural steps from visual evidence alone, pointing to critical gaps in procedural generation.

"Thinking" is inconsistently beneficial. Chain-of-thought reasoning does not uniformly improve performance. For Gemini-2.5-Flash, think mode slightly improves Level-1 (60.2 vs. 58.6) but marginally degrades Level-2 (49.8 vs. 50.1). For MiMo-VL-7B-RL, think mode consistently helps across both levels (44.3 vs. 42.4 on L1; 34.3 vs. 27.4 on L2). This inconsistency suggests that extended reasoning can sometimes cause models to drift from visual evidence toward language-based rationalization, which may undermine procedural scoring.

4.3 Vision-Centric Analysis

Figure 3 shows performance as a function of the number of input frames.

Vision is essential. Removing all visual frames (text-only input) sharply degrades Level-1 and Level-2 performance, and nearly nullifies Level-3 scores. This confirms that ExpVid tasks cannot be solved through language priors alone, validating the vision-centric annotation design.

Frame count and temporal modeling. Long-context video models (e.g., Kimi-VL-A3B-Thinking [Team et al., 2025]) continue to improve as input frames increase up to 256, demonstrating the importance of capturing temporal structure across the full experiment. Image-centric models that process video as a bag of independent frames tend to saturate at lower frame counts, indicating that dedicated temporal modeling is key for advancing performance on ExpVid.

4.4 Discipline-Level Analysis

Performance on ExpVid varies considerably across the 13 scientific disciplines. Biology and chemistry—which are more extensively represented in MLLM pretraining corpora—yield higher overall scores than specialized fields such as neuroscience and plant science. Within disciplines, models show uneven profiles: chemistry experiments with precise volumetric measurements challenge Quantity Recognition (Level-1), whereas neuroscience procedures involving stereotaxic and electrophysiology equipment are particularly difficult for Tool Recognition. At Level-2, medicine experiments with elaborate multi-reagent protocols produce the lowest Step Prediction scores across nearly all models, reflecting both the visual complexity and the domain specificity of clinical preparation workflows. At Level-3, medicine and biochemistry record the lowest Scientific Discovery scores, reflecting the depth of domain knowledge required to interpret clinical or molecular outcomes from visual observations alone. These disciplinary disparities suggest that future work should focus on domain-balanced pretraining and discipline-specific fine-tuning strategies to build models that generalize reliably across the full breadth of scientific experimentation.

AI-Generated

Table 2: Performance of evaluated models on the ExpVid across 10 tasks under three levels.

Model	Think	Level-1					Level-2					Level-3		
		Tool	Mat.	Quan.	Oper.	Avg.	Ord.	Gen.	Veri.	Pred.	Avg.	Anal.	Disc.	Avg.
Human Performance		17.5	15.9	61.3	55.5	37.6	69.8	31.2	45.6	21.8	42.1	–	–	–
Open-source MLLMs														
Qwen2.5-VL-7B-Instruct	×	32.0	33.9	49.0	62.4	42.6	56.2	20.8	20.7	1.3	24.6	25.2	21.4	23.3
MiMo-VL-7B-RL	×	34.2	33.7	44.2	62.4	42.4	43.9	28.5	18.5	11.4	27.4	28.7	25.9	27.3
MiMo-VL-7B-RL	✓	36.1	29.1	53.6	67.8	44.3	64.8	32.3	24.9	15.6	34.3	29.3	27.3	28.3
InternVL3-8B	×	27.5	31.0	38.8	65.6	39.4	43.4	20.4	20.2	3.9	23.9	29.2	25.3	27.2
InternVL3.5-8B	×	27.3	30.8	45.5	64.8	40.3	82.3	25.8	23.7	4.8	34.0	22.6	18.4	20.5
Intern-S1-mini	✓	33.3	31.2	52.5	61.4	42.5	73.6	14.3	16.8	8.3	28.1	33.5	28.3	30.9
Keye-VL-8B-Preview	✓	16.6	22.4	38.9	60.8	32.6	25.4	12.4	19.1	1.7	14.6	9.5	6.7	8.1
Keye-VL-1.5-8B	✓	21.0	23.4	51.3	64.0	37.0	56.7	9.5	20.0	2.8	22.1	8.4	6.1	7.2
GLM-4.1V-9B	✓	30.8	29.8	47.5	59.6	40.1	64.1	18.2	25.0	7.4	28.6	28.1	26.5	27.3
GLM-4.5V	✓	35.5	33.6	61.5	62.3	45.6	71.9	34.9	27.2	12.9	36.6	33.3	32.5	32.9
Kimi-VL-A3B-Thinking	✓	34.6	32.6	40.7	59.5	40.8	32.3	18.2	23.3	6.2	20.0	24.6	21.8	23.2
InternVL3.5-38B	✓	35.9	34.0	46.7	65.3	44.0	65.8	36.7	23.0	19.0	36.0	33.1	30.8	31.9
InternVL3-78B	✓	35.1	34.3	**73.2**	75.8	50.9	**87.1**	45.5	19.8	15.5	41.9	40.3	35.3	37.7
Qwen2.5-VL-72B-Instruct	×	30.5	34.7	54.5	64.5	43.9	86.3	34.1	23.8	0.3	35.9	31.9	29.3	30.6
Intern-S1	✓	38.9	35.2	58.9	73.8	49.9	82.2	45.0	24.1	15.4	36.0	43.0	36.3	39.6
Closed-source MLLMs														
Seed-VL-1.5	✓	32.9	24.6	43.9	69.2	40.7	73.9	48.6	19.8	27.9	42.5	32.0	29.4	30.7
Claude-Sonnet-4	×	25.6	31.2	54.3	61.9	40.8	78.7	37.6	16.5	11.6	36.0	29.1	30.1	29.6
Gemini-2.5-Flash	×	52.7	50.1	65.2	72.6	58.6	86.0	50.5	24.1	40.2	50.1	47.2	41.1	44.1
Gemini-2.5-Flash	✓	52.7	**50.7**	71.9	73.3	**60.2**	85.1	54.3	22.3	38.0	49.8	44.8	41.3	43.0
Gemini-2.5-Pro	×	**53.1**	45.9	64.3	**80.8**	59.2	83.7	61.3	**26.8**	49.6	53.8	50.6	45.2	47.9
Gemini-2.5-Pro	✓	51.3	44.3	63.8	74.4	56.7	84.2	59.9	**26.8**	46.9	54.3	50.1	44.8	47.4
GPT-5	✓	51.6	37.8	59.5	71.9	53.3	85.1	**66.9**	**26.8**	**51.8**	**57.5**	**55.4**	**57.4**	**56.4**

AI-Generated

5 Conclusion

We introduced ExpVid, a benchmark for evaluating MLLMs on authentic, peer-reviewed laboratory experiment videos. By organizing 10 tasks into a three-level hierarchy—visual perception, procedural understanding, and scientific reasoning—ExpVid challenges models to exhibit fine-grained procedural knowledge alongside high-order scientific inference. Built from 390 JoVE videos spanning 13 disciplines, the benchmark yields 7,800 vision-centric QA pairs with rigorous PhD-level expert validation.

Our comprehensive evaluation of 19 MLLMs reveals four key findings: (1) closed-source frontier models substantially outperform open-source counterparts at all three levels; (2) all models struggle profoundly with procedural generation and prediction, exposing a critical gap in temporal generative reasoning; (3) chain-of-thought reasoning offers inconsistent benefits and can occasionally degrade procedural accuracy; and (4) visual information is indispensable, especially for Level-3 scientific reasoning tasks. Together, these findings define concrete directions for future work: richer temporal modeling, stronger visual grounding of reasoning chains, and training on procedurally structured scientific data.

We hope ExpVid serves both as a diagnostic tool and as a roadmap for advancing multimodal scientific assistants—systems capable of perceiving, verifying, and reasoning about the real laboratory workflows that underpin scientific discovery.

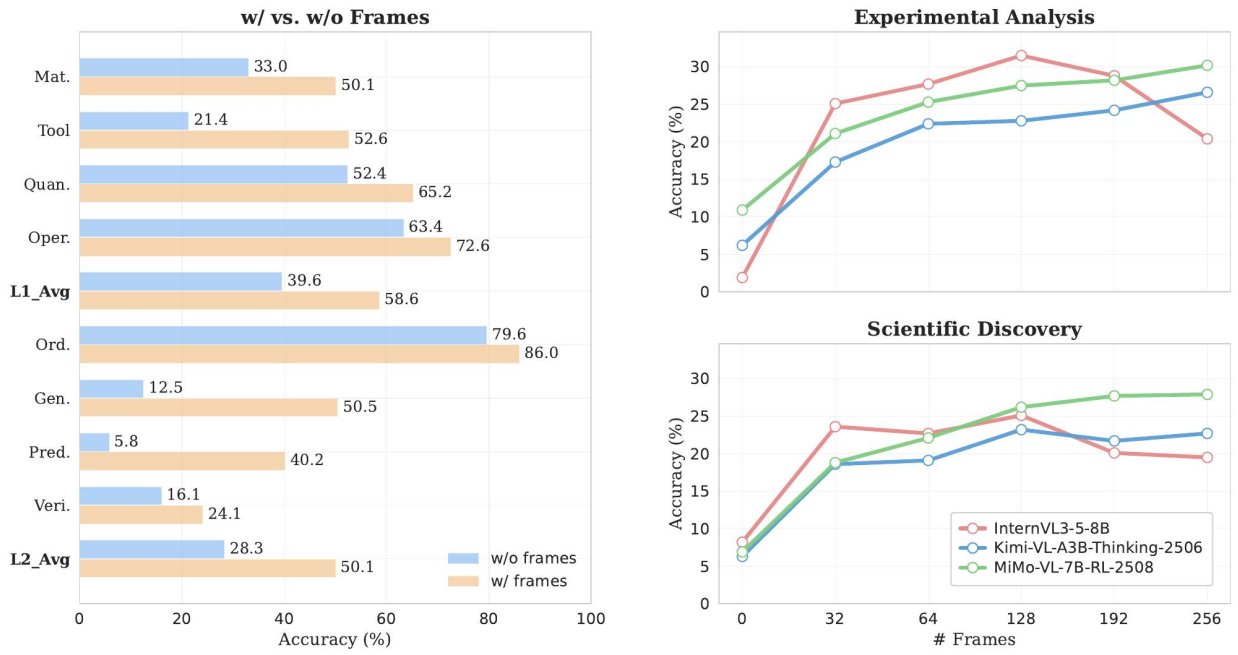

Figure 3: Effect of input video frames on ExpVid performance. Removing all frames drastically degrades performance, particularly on Level-3. Long-context video models benefit from additional frames, while image-centric models saturate earlier.

AI-Generated

References

Marah Abdin, Jyoti Aneja, Hany Awadalla, and . . . . Phi-3 technical report: A highly capable language model locally on your phone. *arXiv preprint arXiv:2404.14219*, 2024.

Anthropic. Claude sonnet 4. `https://www.anthropic.com/claude/sonnet`, 2025. Accessed: 2025-09-21.

Lei Bai, Zhongrui Cai, Maosong Cao, Weihan Cao, Chiyu Chen, Haojiong Chen, Kai Chen, Pengcheng Chen, Ying Chen, Yongkang Chen, et al. Intern-s1: A scientific multimodal foundation model. *arXiv preprint arXiv:2508.15763*, 2025a.

Shuai Bai, Keqin Chen, Xuejing Liu, Jialin Wang, Wenbin Ge, Sibo Song, Kai Dang, Peng Wang, Shijie Wang, Jun Tang, et al. Qwen2. 5-vl technical report. *arXiv preprint arXiv:2502.13923*, 2025b.

Fabian Caba Heilbron, Victor Escorcia, Bernard Ghanem, and Juan Carlos Niebles. Activitynet: A large-scale video benchmark for human activity understanding. In *Proceedings of the ieee conference on computer vision and pattern recognition*, pages 961–970, 2015.

Lin Chen, Jinsong Li, Xiaoyi Dong, Pan Zhang, Yuhang Zang, Zehui Chen, Haodong Duan, Jiaqi Wang, Yu Qiao, Dahua Lin, et al. Are we on the right way for evaluating large vision-language models? *Advances in Neural Information Processing Systems*, 37:27056–27087, 2024.

Pradipto Das, Chenliang Xu, Richard F Doell, and Jason J Corso. A thousand frames in just a few words: Lingual description of videos through latent topics and sparse object stitching. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 2634–2641, 2013.

Google DeepMind. Gemini 2.5 flash. `https://deepmind.google/models/gemini/flash/`, 2025a. Accessed: 2025-09-24.

Google DeepMind. Gemini 2.5 pro. `https://deepmind.google/technologies/gemini`, 2025b. Accessed: 2025-09-21.

Chaoyou Fu, Yuhan Dai, Yongdong Luo, Lei Li, Shuhuai Ren, Renrui Zhang, Zihan Wang, Chenyu Zhou, Yunhang Shen, Mengdan Zhang, et al. Video-mme: The first-ever comprehensive evaluation benchmark of multi-modal llms in video analysis. In *Proceedings of the Computer Vision and Pattern Recognition Conference*, pages 24108–24118, 2025.

Gianmarco Gabrieli, Irina Espejo Morales, Dimitrios Christofidellis, Mara Graziani, Andrea Giovannini, Federico Zipoli, Amol Thakkar, Antonio Foncubierta, Matteo Manica, and Patrick W Ruch. Activity recognition in scientific experimentation using multimodal visual encoding. *Digital Discovery*, 4(2):393–402, 2025.

Daya Guo, Dejian Yang, Haowei Zhang, Junxiao Song, Ruoyu Zhang, Runxin Xu, Qihao Zhu, Shirong Ma, Peiyi Wang, Xiao Bi, et al. Deepseek-r1: Incentivizing reasoning capability in llms via reinforcement learning. *arXiv preprint arXiv:2501.12948*, 2025a.

Dong Guo, Faming Wu, Feida Zhu, Fuxing Leng, Guang Shi, Haobin Chen, Haoqi Fan, Jian Wang, Jianyu Jiang, Jiawei Wang, et al. Seed1. 5-vl technical report. *arXiv preprint arXiv:2505.07062*, 2025b.

Yana Hasson, Pauline Luc, Liliane Momeni, Maks Ovsjanikov, Guillaume Le Moing, Alina Kuznetsova, Ira Ktena, Jennifer J Sun, Skanda Koppula, Dilara Gokay, et al. Scivid: Cross-domain evaluation of video models in scientific applications. *arXiv preprint arXiv:2507.03578*, 2025.

Wenyi Hong, Wenmeng Yu, Xiaotao Gu, Guo Wang, Guobing Gan, Haomiao Tang, Jiale Cheng, Ji Qi, Junhui Ji, Lihang Pan, et al. Glm-4.1 v-thinking: Towards versatile multimodal reasoning with scalable reinforcement learning. *arXiv e-prints*, pages arXiv–2507, 2025.

Kairui Hu, Penghao Wu, Fanyi Pu, Wang Xiao, Yuanhan Zhang, Xiang Yue, Bo Li, and Ziwei Liu. Video-mmmu: Evaluating knowledge acquisition from multi-discipline professional videos. *arXiv preprint arXiv:2501.13826*, 2025.

JoVE. Jove: Journal of visualized experiments. `https://www.jove.com`. Accessed: 2025-09-12.

Jie Lei, Tamara L Berg, and Mohit Bansal. Detecting moments and highlights in videos via natural language queries. *Advances in Neural Information Processing Systems*, 34:11846–11858, 2021.

Kunchang Li, Yali Wang, Yinan He, Yizhuo Li, Yi Wang, Yi Liu, Zun Wang, Jilan Xu, Guo Chen, Ping Luo, et al. Mvbench: A comprehensive multi-modal video understanding benchmark. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 22195–22206, 2024.

Pan Lu, Hritik Bansal, Tony Xia, Jiacheng Liu, Chunyuan Li, Hannaneh Hajishirzi, Hao Cheng, Kai-Wei Chang, Michel Galley, and Jianfeng Gao. Mathvista: Evaluating mathematical reasoning of foundation models in visual contexts. *arXiv preprint arXiv:2310.02255*, 2023.

Karttikeya Mangalam, Raiymbek Akshulakov, and Jitendra Malik. Egoschema: A diagnostic benchmark for very long-form video language understanding. *Advances in Neural Information Processing Systems*, 36:46212–46244, 2023.

OpenAI. Gpt-5. `https://openai.com/gpt-5`. 2025. Accessed: 2025-09-21.

Tanik Saikh, Tirthankar Ghosal, Amish Mittal, Asif Ekbal, and Pushpak Bhattacharyya. Scienceqa: A novel resource for question answering on scholarly articles. *International Journal on Digital Libraries*, 23(3):289–301, 2022.

Gunnar A Sigurdsson, Gül Varol, Xiaolong Wang, Ali Farhadi, Ivan Laptev, and Abhinav Gupta. Hollywood in homes: Crowdsourcing data collection for activity understanding. In *European conference on computer vision*, pages 510–526. Springer, 2016.

Kimi Team, Angang Du, Bohong Yin, Bowei Xing, Bowen Qu, Bowen Wang, Cheng Chen, Chenlin Zhang, Chenzhuang Du, Chu Wei, et al. Kimi-vl technical report. *arXiv preprint arXiv:2504.07491*, 2025.

Bin Wang, Chao Xu, Xiaomeng Zhao, Linke Ouyang, Fan Wu, Zhiyuan Zhao, Rui Xu, Kaiwen Liu, Yuan Qu, Fukai Shang, et al. Mineru: An open-source solution for precise document content extraction. *arXiv preprint arXiv:2409.18839*, 2024a.

Ke Wang, Junting Pan, Weikang Shi, Zimu Lu, Houxing Ren, Aojun Zhou, Mingjie Zhan, and Hongsheng Li. Measuring multimodal mathematical reasoning with math-vision dataset. *Advances in Neural Information Processing Systems*, 37:95095–95169, 2024b.

Weiyun Wang, Zhangwei Gao, Lixin Gu, Hengjun Pu, Long Cui, Xingguang Wei, Zhaoyang Liu, Linglin Jing, Shenglong Ye, Jie Shao, et al. Internvl3. 5: Advancing open-source multimodal models in versatility, reasoning, and efficiency. *arXiv preprint arXiv:2508.18265*, 2025.

Xiaoxuan Wang, Ziniu Hu, Pan Lu, Yanqiao Zhu, Jieyu Zhang, Satyen Subramaniam, Arjun R Loomba, Shichang Zhang, Yizhou Sun, and Wei Wang. Scibench: Evaluating college-level scientific problem-solving abilities of large language models. *arXiv preprint arXiv:2307.10635*, 2023.

Takuma Yagi, Misaki Ohashi, Yifei Huang, Ryosuke Furuta, Shungo Adachi, Toutai Mitsuyama, and Yoichi Sato. Finebio: A fine-grained video dataset of biological experiments with hierarchical annotation. *International Journal of Computer Vision*, pages 1–16, 2025.

AI-Generated

Jiashuo Yu, Yue Wu, Meng Chu, Zhifei Ren, Zizheng Huang, Pei Chu, Ruijie Zhang, Yinan He, Qirui Li, Songze Li, et al. Vrbench: A benchmark for multi-step reasoning in long narrative videos. *arXiv preprint arXiv:2506.10857*, 2025.

Xiang Yue, Yuansheng Ni, Kai Zhang, Tianyu Zheng, Ruoqi Liu, Ge Zhang, Samuel Stevens, Dongfu Jiang, Weiming Ren, Yuxuan Sun, et al. Mmmu: A massive multi-discipline multimodal understanding and reasoning benchmark for expert agi. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 9556–9567, 2024.

Yilun Zhao, Haowei Zhang, Lujing Xie, Tongyan Hu, Guo Gan, Yitao Long, Zhiyuan Hu, Weiyuan Chen, Chuhan Li, Zhijian Xu, et al. Mmvu: Measuring expert-level multi-discipline video understanding. In *Proceedings of the Computer Vision and Pattern Recognition Conference*, pages 8475–8489, 2025.

Junjie Zhou, Yan Shu, Bo Zhao, Boya Wu, Shitao Xiao, Xi Yang, Yongping Xiong, Bo Zhang, Tiejun Huang, and Zheng Liu. Mlvu: A comprehensive benchmark for multi-task long video understanding. *arXiv e-prints*, pages arXiv–2406, 2024.

Yuhao Zhou, Yiheng Wang, Xuming He, Ruoyao Xiao, Zhiwei Li, Qiantai Feng, Zijie Guo, Yuejin Yang, Hao Wu, Wenxuan Huang, et al. Scientists' first exam: Probing cognitive abilities of mllm via perception, understanding, and reasoning. *arXiv preprint arXiv:2506.10521*, 2025.

Jinguo Zhu, W Wang, Z Chen, Z Liu, S Ye, L Gu, H Tian, Y Duan, W Su, J Shao, et al. Internvl3: Exploring advanced training and test-time recipes for open-source multimodal models, 2025. *arXiv preprint arXiv:2504.10479*, 9.

AI-Generated